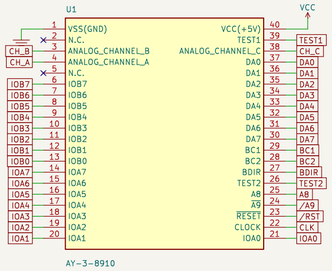
PIN CONFIGURATION

40 lead dual in line, top view



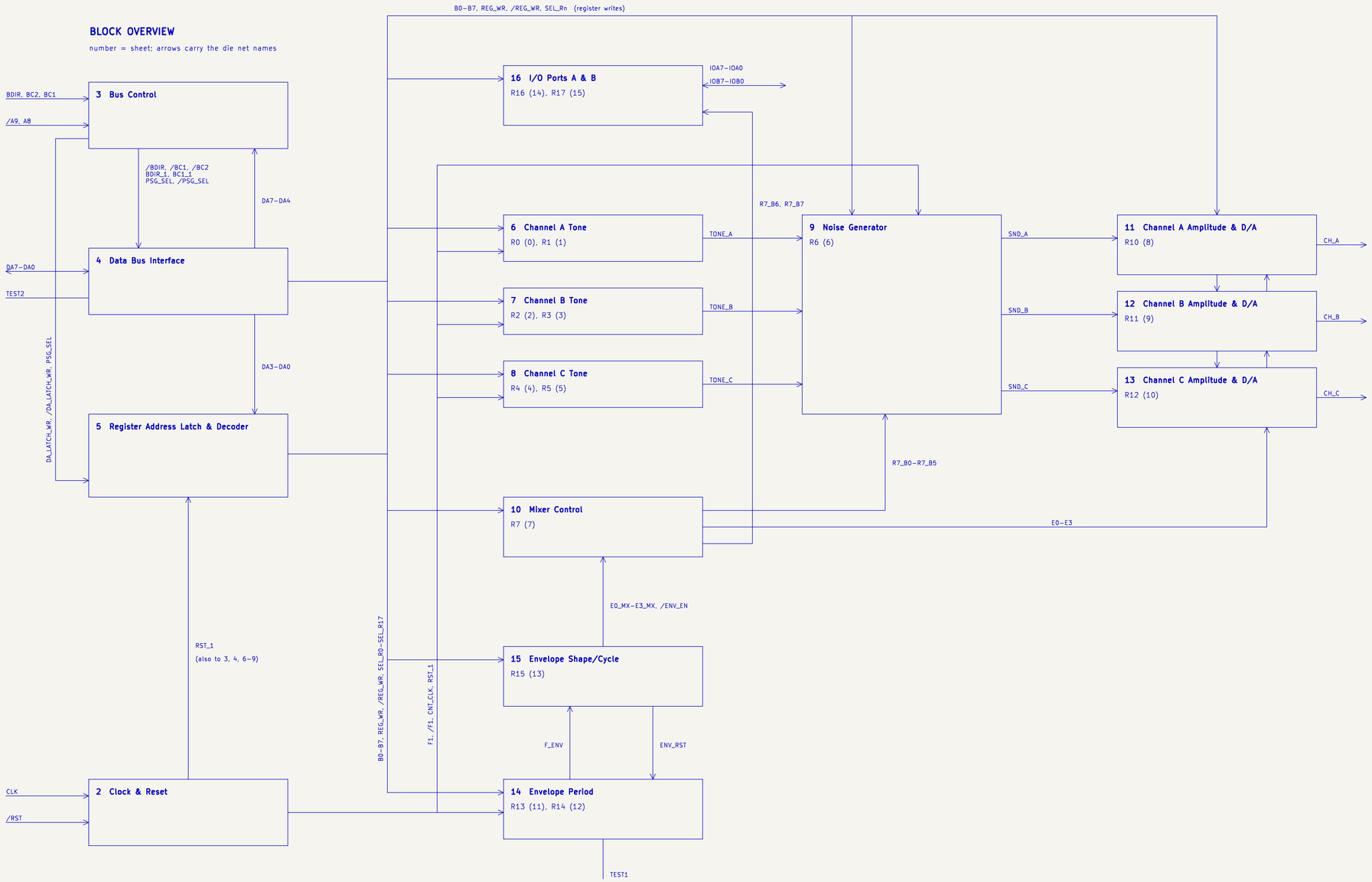
REGISTER ARRAY

Register	Function	Sheet
R0 (0)	Channel A Tone Period, fine	6
R1 (1)	Channel A Tone Period, coarse	6
R2 (2)	Channel B Tone Period, fine	7
R3 (3)	Channel B Tone Period, coarse	7
R4 (4)	Channel C Tone Period, fine	8
R5 (5)	Channel C Tone Period, coarse	8
R6 (6)	Noise Period	9
R7 (7)	Enable (tone, noise, I/O direction)	10
R10 (8)	Channel A Amplitude	11
R11 (9)	Channel B Amplitude	12
R12 (10)	Channel C Amplitude	13
R13 (11)	Envelope Period, fine	14
R14 (12)	Envelope Period, coarse	14
R15 (13)	Envelope Shape/Cycle	15
R16 (14)	I/O Port A Data Store	16
R17 (15)	I/O Port B Data Store	16

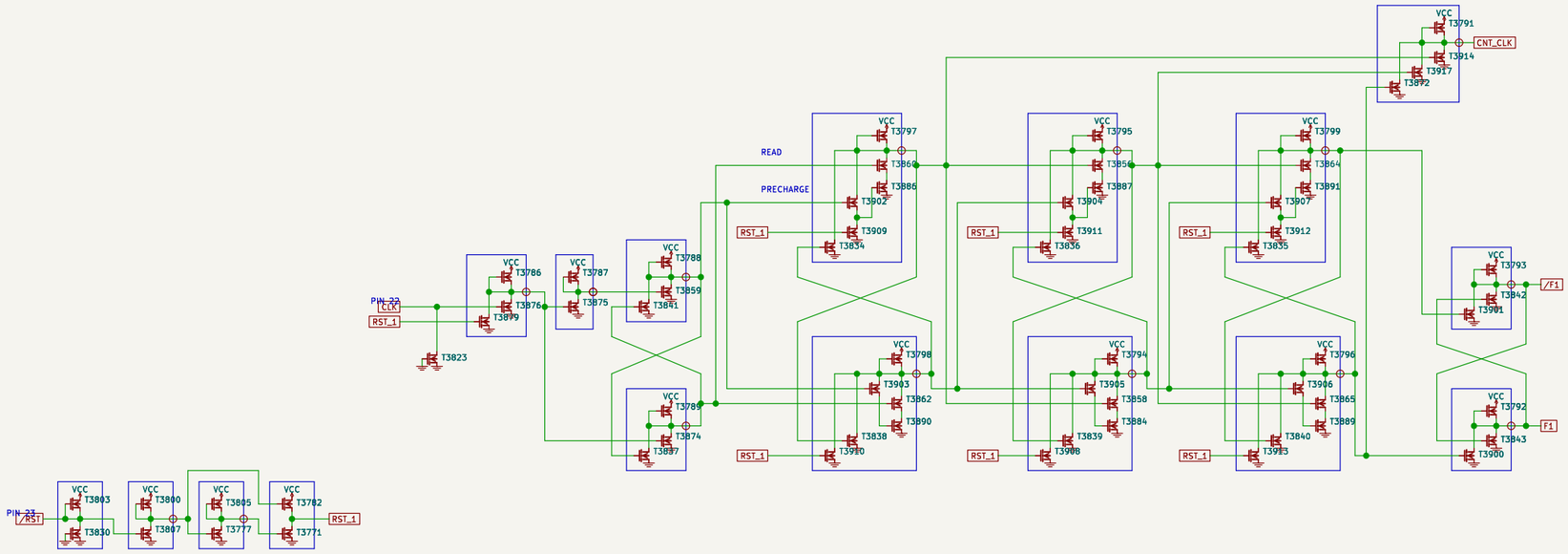
Register numbers are octal as in the GI data manual, decimal in brackets.

BLOCK OVERVIEW

number = sheet; arrows carry the die net names

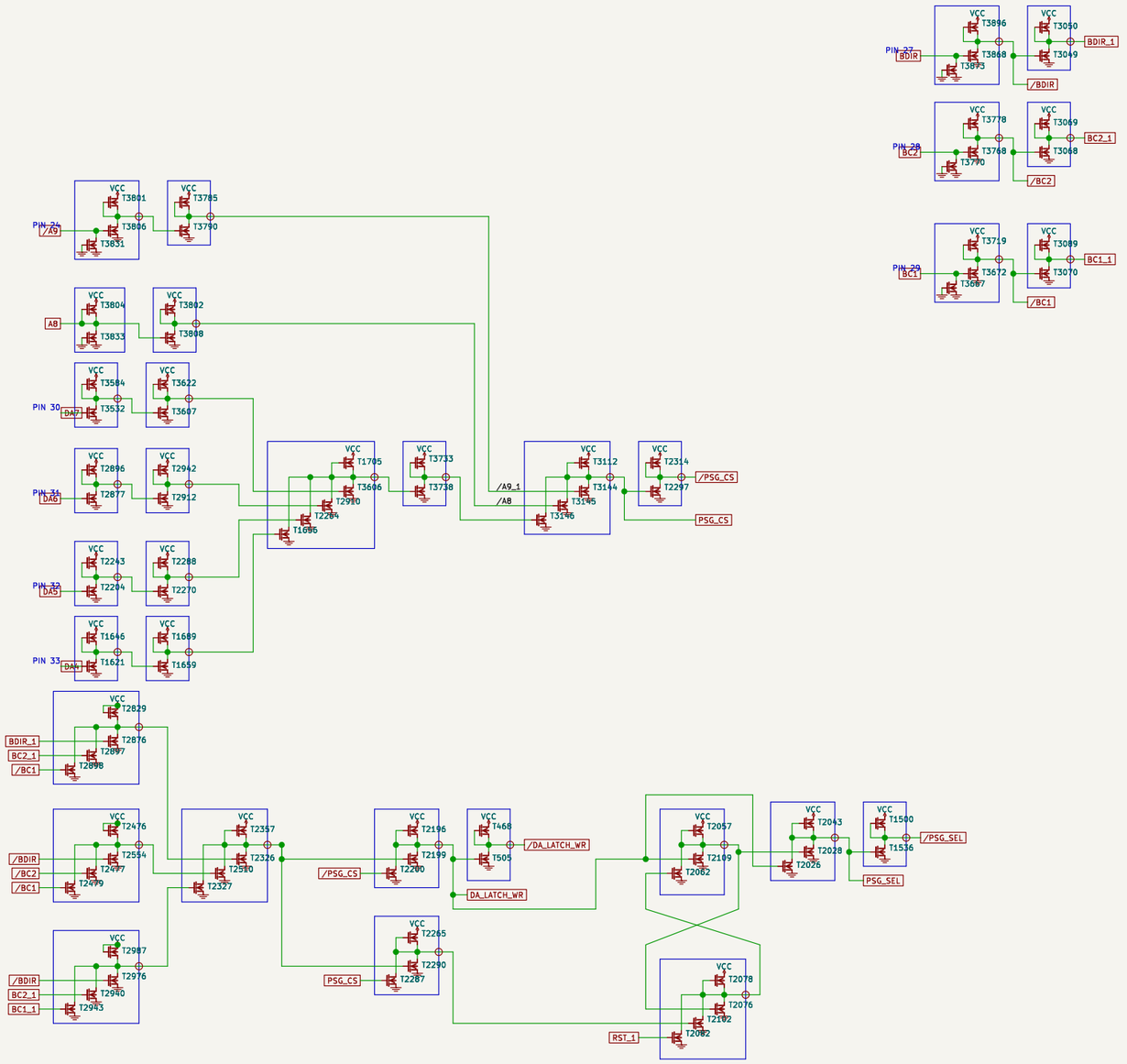


Die:	General Instrument AY-3-8910 (1979)
Sheet:	1 / 16
File:	ay8910_ctrl_s1.kicad_sch
Title: Pinout & Block Overview	
Size: A2	Date: 2026-10-09
KiCad E.D.A. 9.0.1	Rev: 1.0
	Id: 1/1



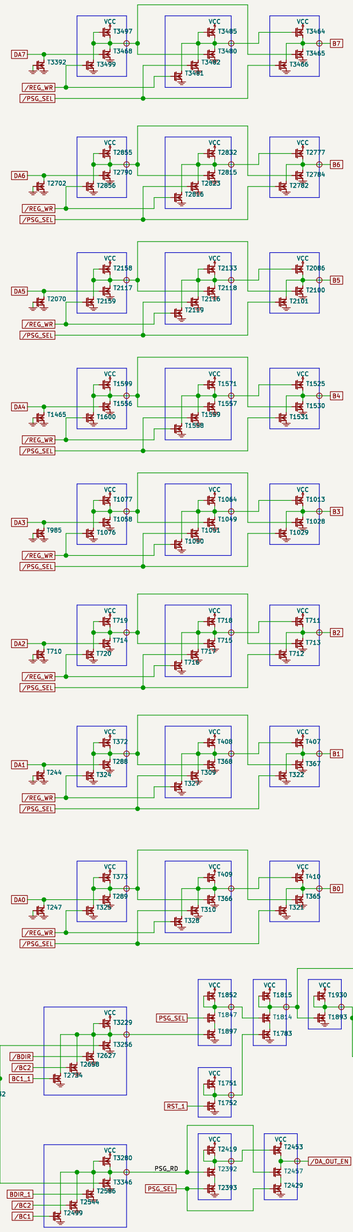
Die: General Instrument AY-3-8910 (1979)	
Sheet: 2 / 16 File: ay8910_ctrl_s2.kicad_sch	
Title: Clock & Reset	
Size: A3	Date: 2026-10-09
KiCad E.D.A. 9.0.1	Id: 1/1

Rev: 1.0

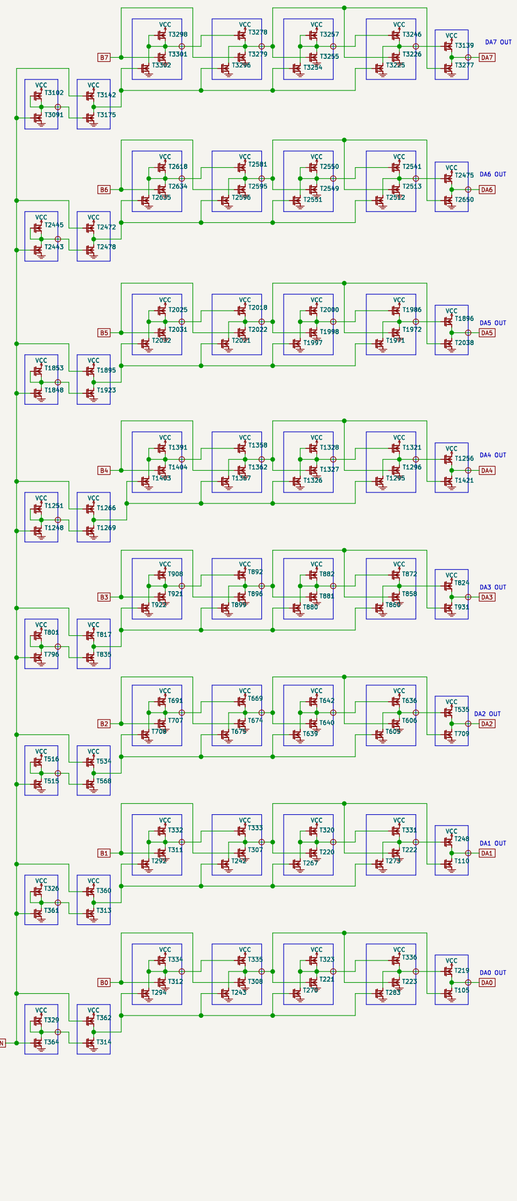


Die: General Instrument AY-3-8910 (1979)		
Sheet: 3 / 16 File: ay8910_ctrl_s3.kicad_sch		
Title: Bus Control		
Size: A3	Date: 2026-10-09	**Rev: 1.0**
KiCad E.D.A. 9.0.1		Id: 1/1

DA->B L

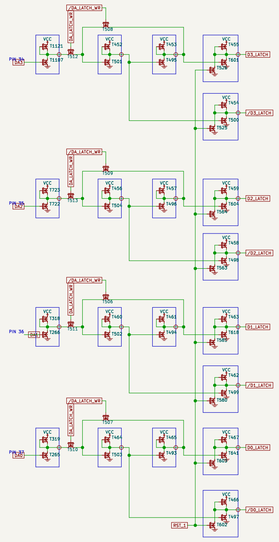


B->DA A

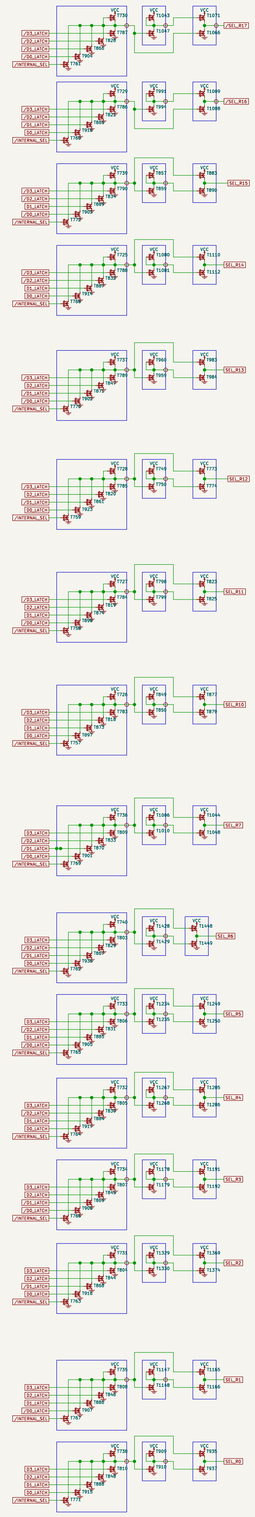


Die: General Instrument AY-3-8910 (1979)		
Sheet: 4 / 16		
File: ay8910_ctrl_s4.kicad_sch		
Title: Data Bus Interface		
Size: A2	Date: 2026-10-09	**Rev: 1.0**
KiCad E.D.A. 9.0.1		Id: 1/1

REG ADDR LATCH



REG ADDR DECODER

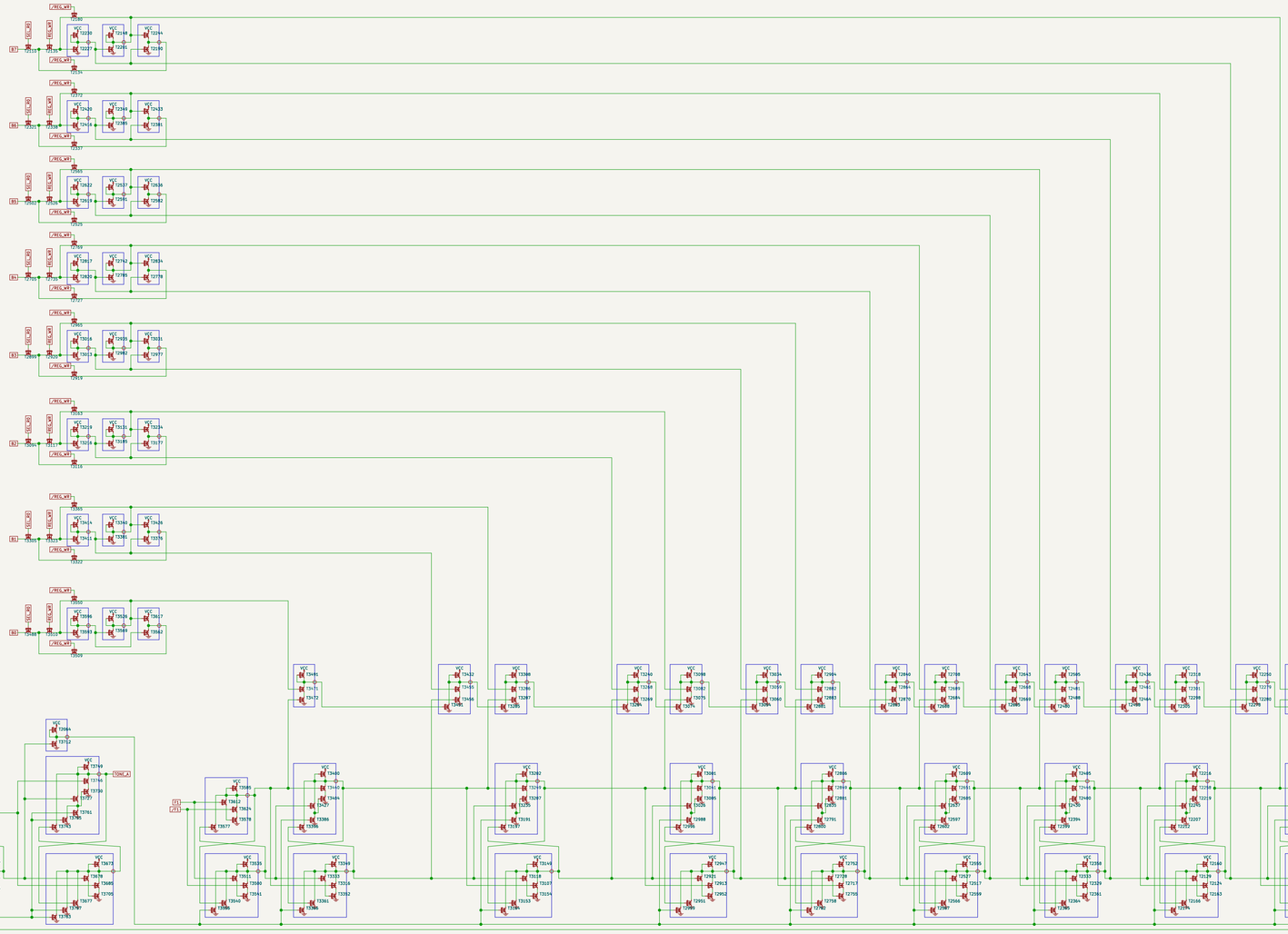




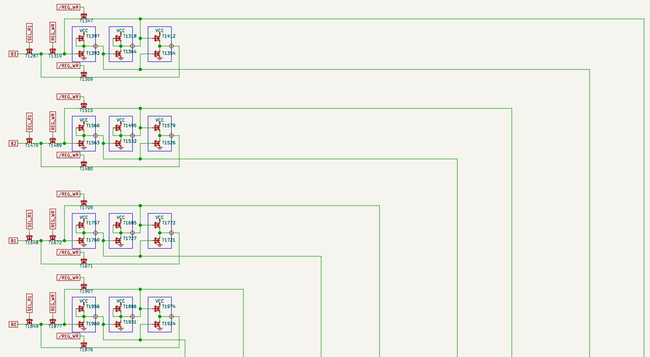
Die: General Instrument AY-3-8910 (1979)	
Sheet: 5 / 16	
File: ay8910_ctrl_s5.kicad_sch	
Title: Register Address Latch & Decoder	
Size: A1	Date: 2026-10-09
KiCad E.D.A. 9.0.1	Id: 1/1

CH_A TONE PERIOD

R0 (0)



R1 (1)



Die: General Instrument AY-3-8910 (1979)	
Sheet: 6 / 16	
File: ay8910_tone_a6.kicad_sch	
Title: Channel A Tone — R0 (0), R1 (1)	
Size: A1	Date: 2026-10-09
KiCad E.D.A. 9.0.1	**Rev: 1.0**
	Id: 1/1

CH_B TONE PERIOD

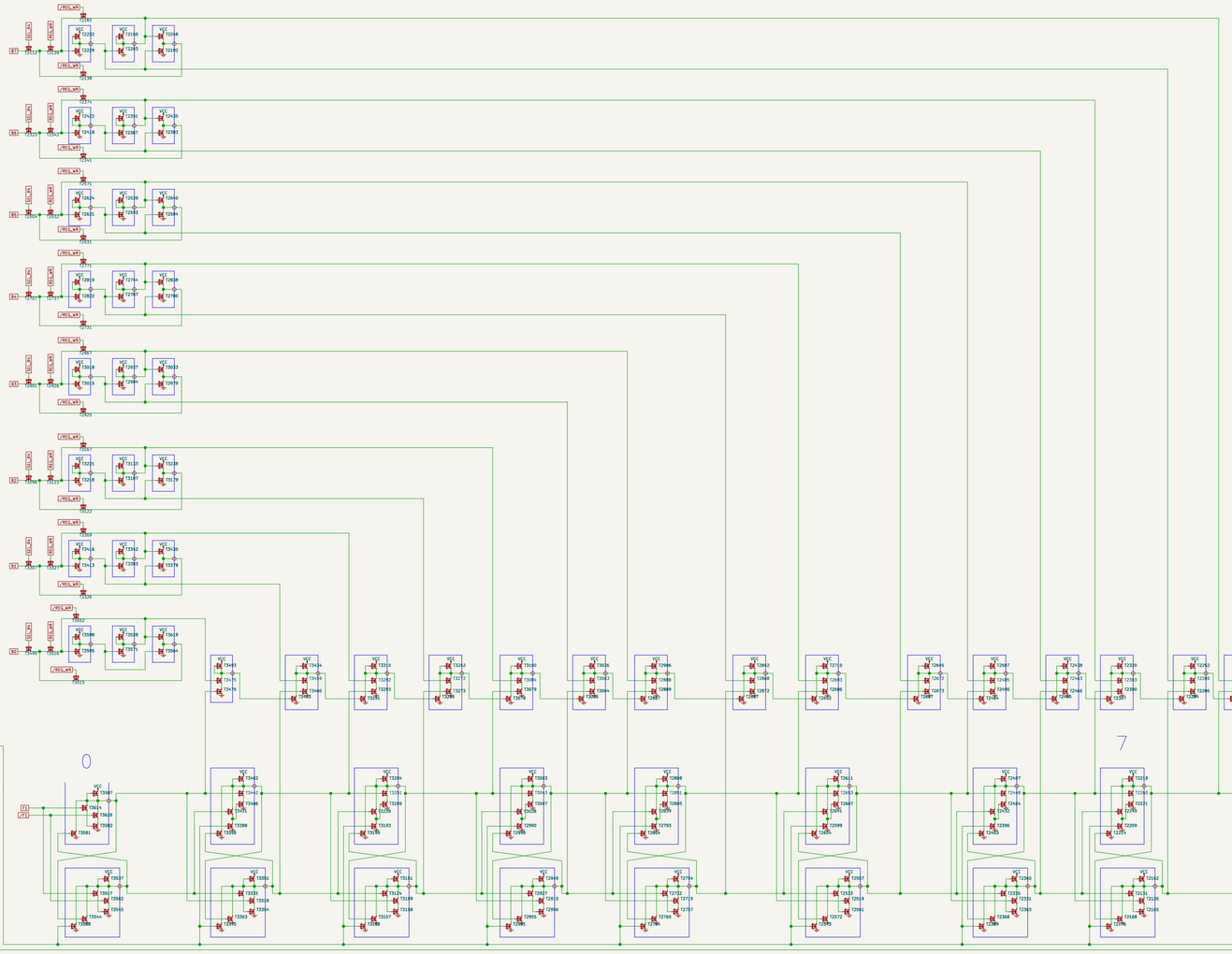
R2 (2)

R3 (3)

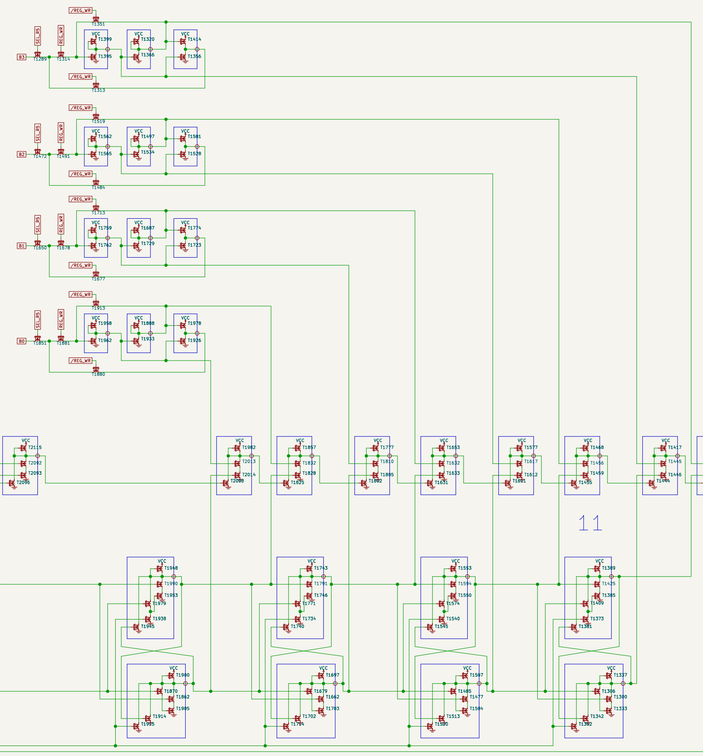
File:	ay8910_tone_s7.kicad_sch
Title:	Channel B Tone — R2 (2), R3 (3)
Size:	A1
Date:	2026-10-09
Rev:	1.0
KiCad E.D.A.	9.0.1
Id:	1/1

CH_C TONE PERIOD

R4 (4)



R5 (5)

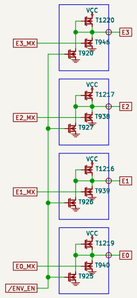


Die: General instrument AY-3-8910 (1979)	
Sheet: 8 / 16	
File: ay8910_tone_x8.kicad_sch	
Title: Channel C Tone — R4 (4), R5 (5)	
Size: A1	Date: 2026-10-09
KiCad E.D.A. 9.0.1	**Rev: 1.0**
	Id: 1/1

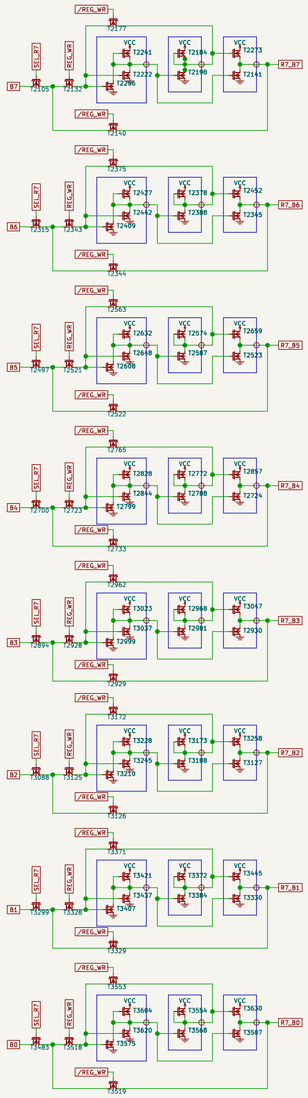
NOISE GEN

R6 (6) NOISE PERIOD

Doc:	General Instrument AY-3-8910 (1979)
Sheet:	9 / 16
File:	ay8910_noise_v9.kicad_sch
Title:	**Noise Generator — R6 (6)**
Size:	A0
Date:	2026-10-09
Rev:	1.0
KiCad E.D.A.	9.0.1
Id:	1/1

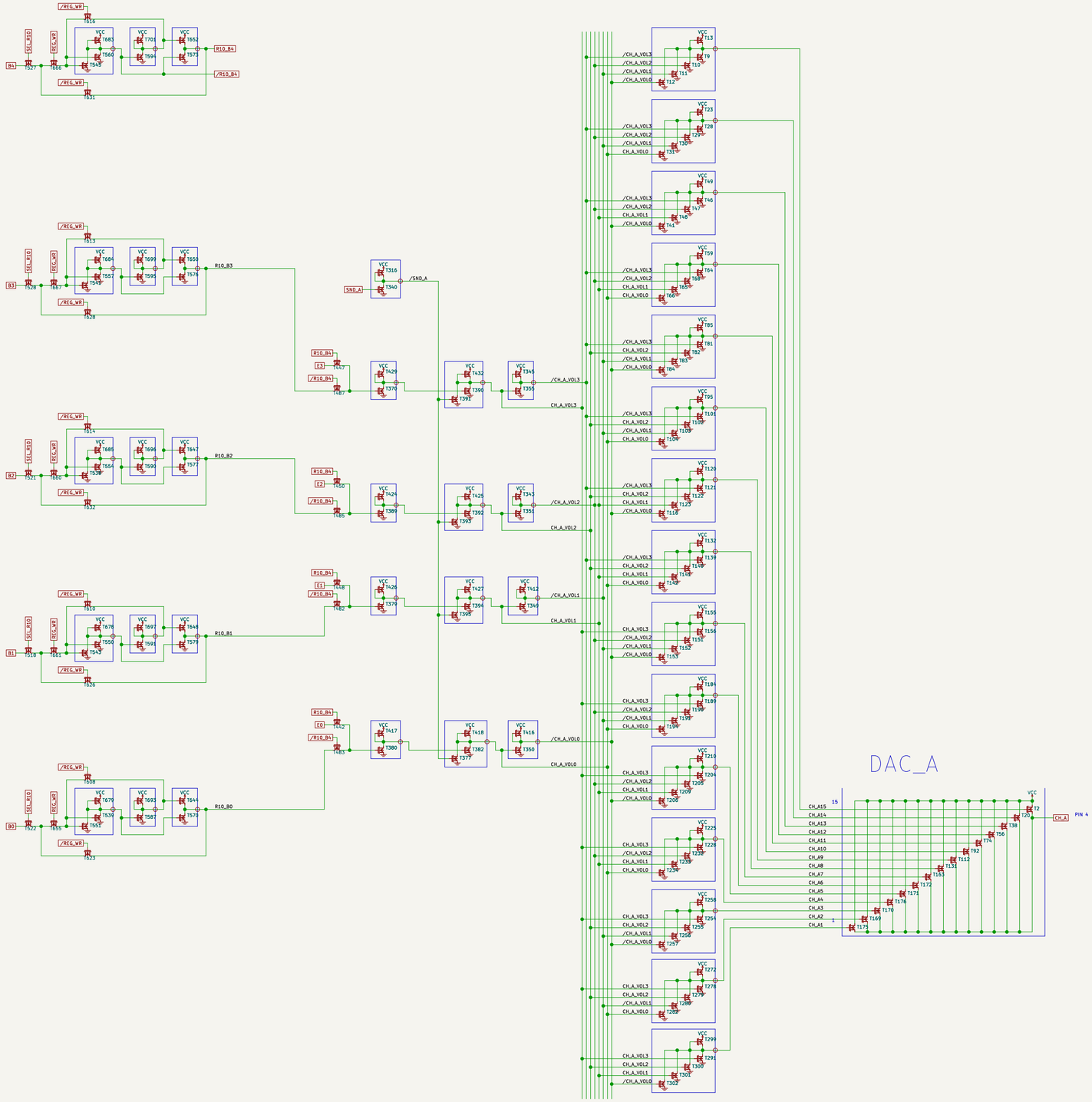


R7 (7) MIXER CONTROL



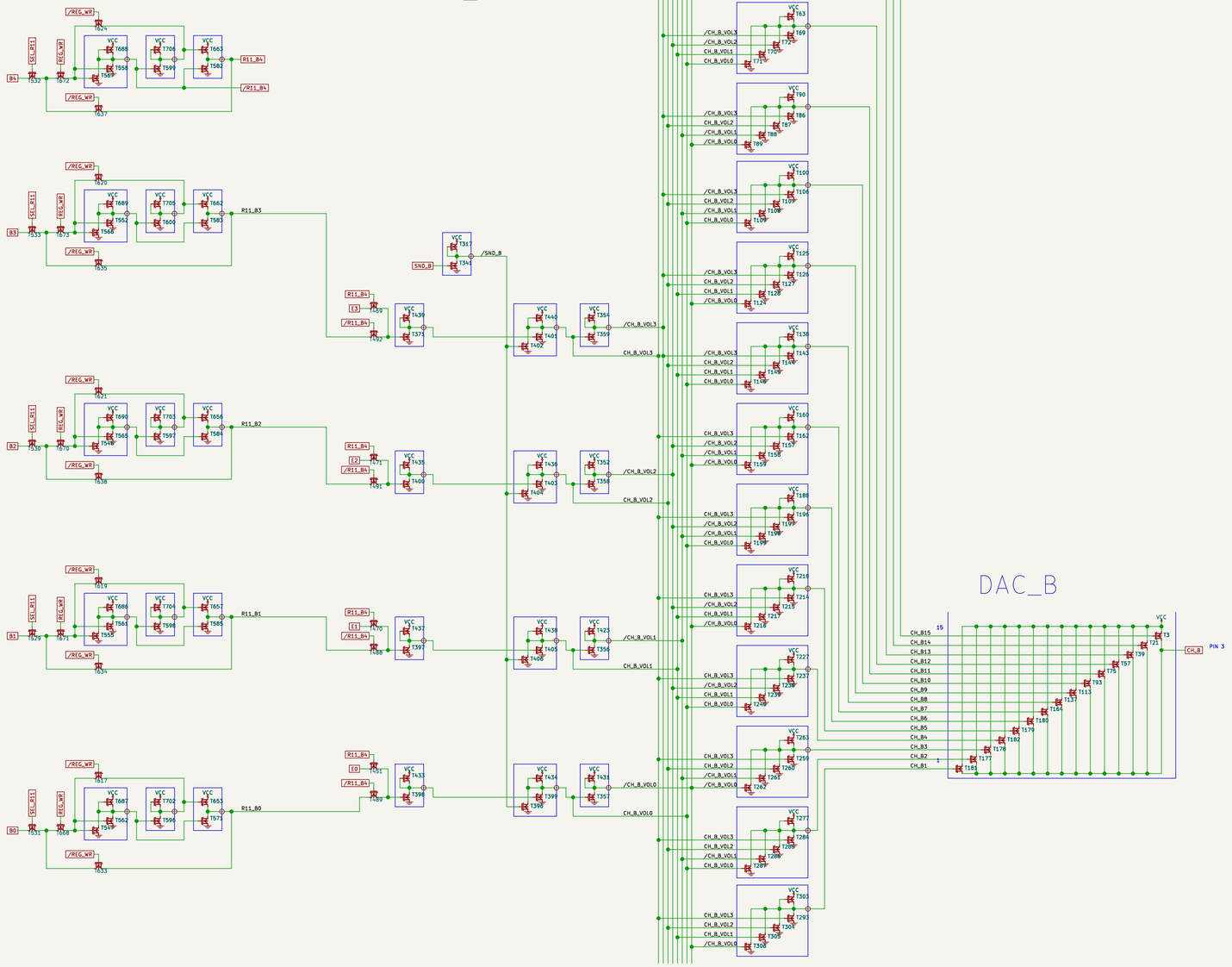
Die: General Instrument AY-3-8910 (1979)		
Sheet: 10 / 16 File: ay8910_noise_s10.kicad_sch		
Title: Mixer Control – R7 (7)		
Size: A2	Date: 2026-10-09	**Rev: 1.0**
KiCad E.D.A. 9.0.1		Id: 1/1

R10 (8) CH_A AMPLITUDE



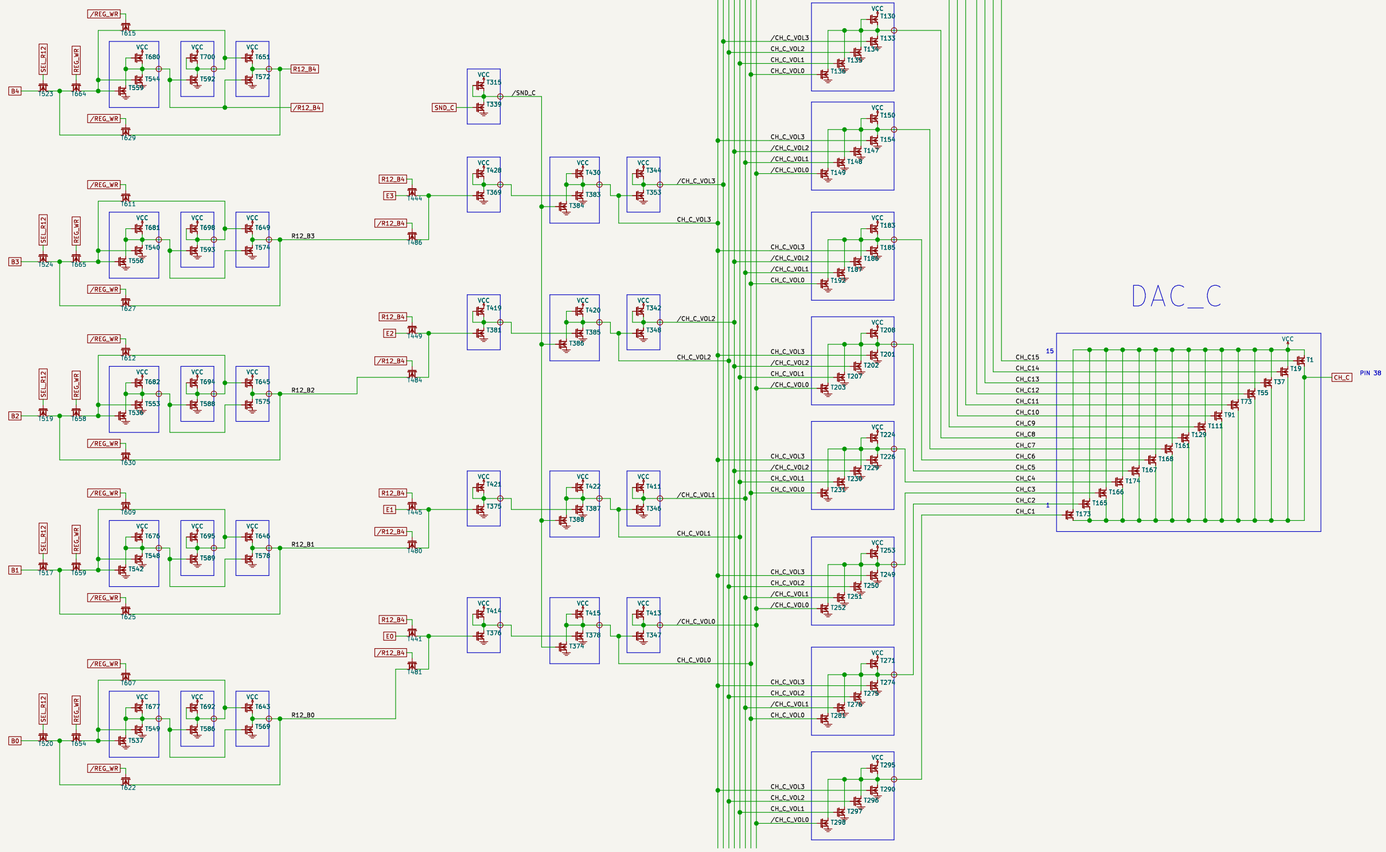
Die:	General Instrument AY-3-8910 (1979)
Sheet:	11 / 16
File:	ay8910_amp_s11.kicad_sch
Title:	***Channel A Amplitude & D/A – R10 (8)***
Size:	A2
Date:	2026-10-09
Rev:	**1.0**
KiCad E.D.A.	9.0.1
Id:	1/1

R11 (9) CH_B AMPLITUDE



Die:	General Instrument AY-3-8910 (1979)
Sheet:	12 / 16
File:	ay8910_amp_s12.kicad_sch
Title: Channel B Amplitude & D/A – R11 (9)	
Size: A2	Date: 2026-10-09
KiCad E.D.A. 9.0.1	Id: 1/1
	Rev: 1.0

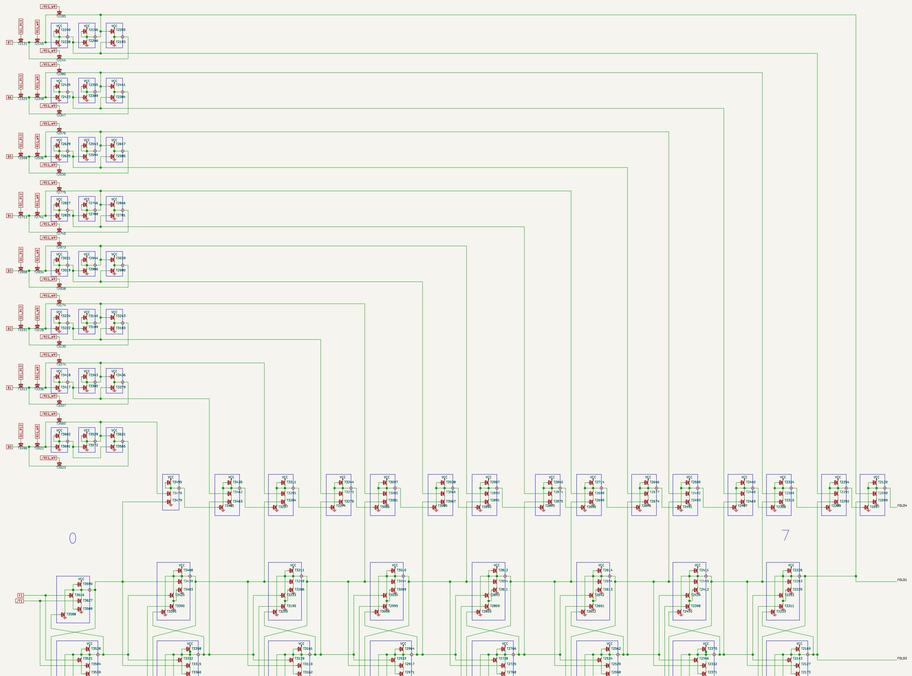
R12 (10) CH_C AMPLITUDE



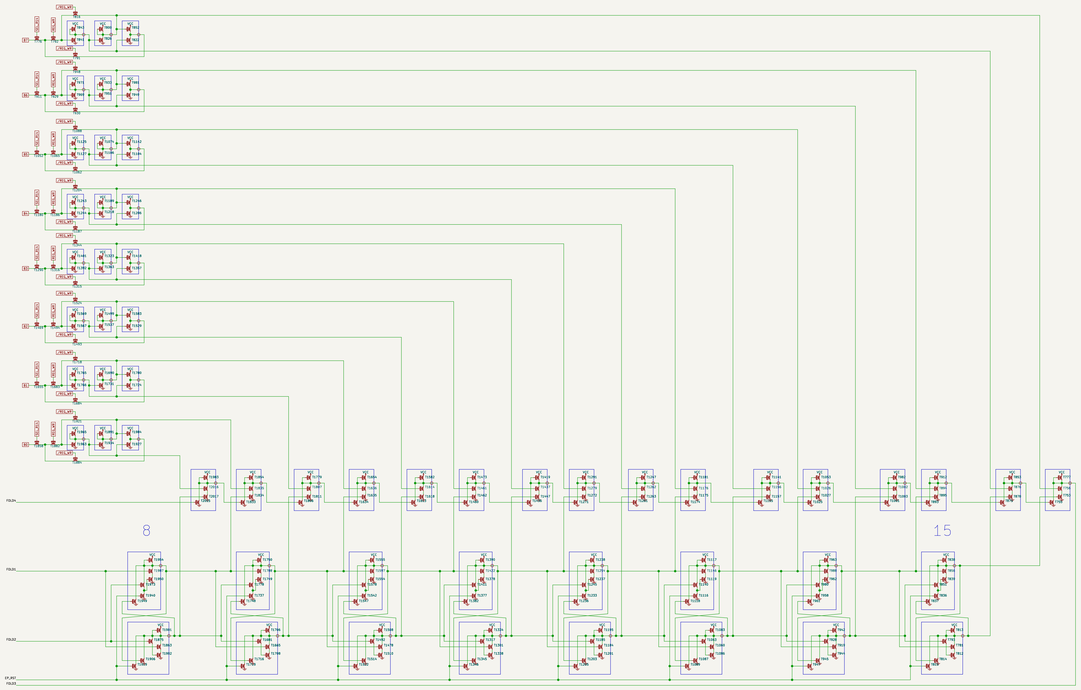
Die:	General Instrument AY-3-8910 (1979)
Sheet:	13 / 16
File:	ay8910_amp_s13.kicad_sch
Title:	***Channel C Amplitude & D/A – R12 (10)***
Size: A2	Date: 2026-10-09
KiCad E.D.A. 9.0.1	**Rev: 1.0**
	Id: 1/1

ENVELOPE PERIOD

R13 (11)

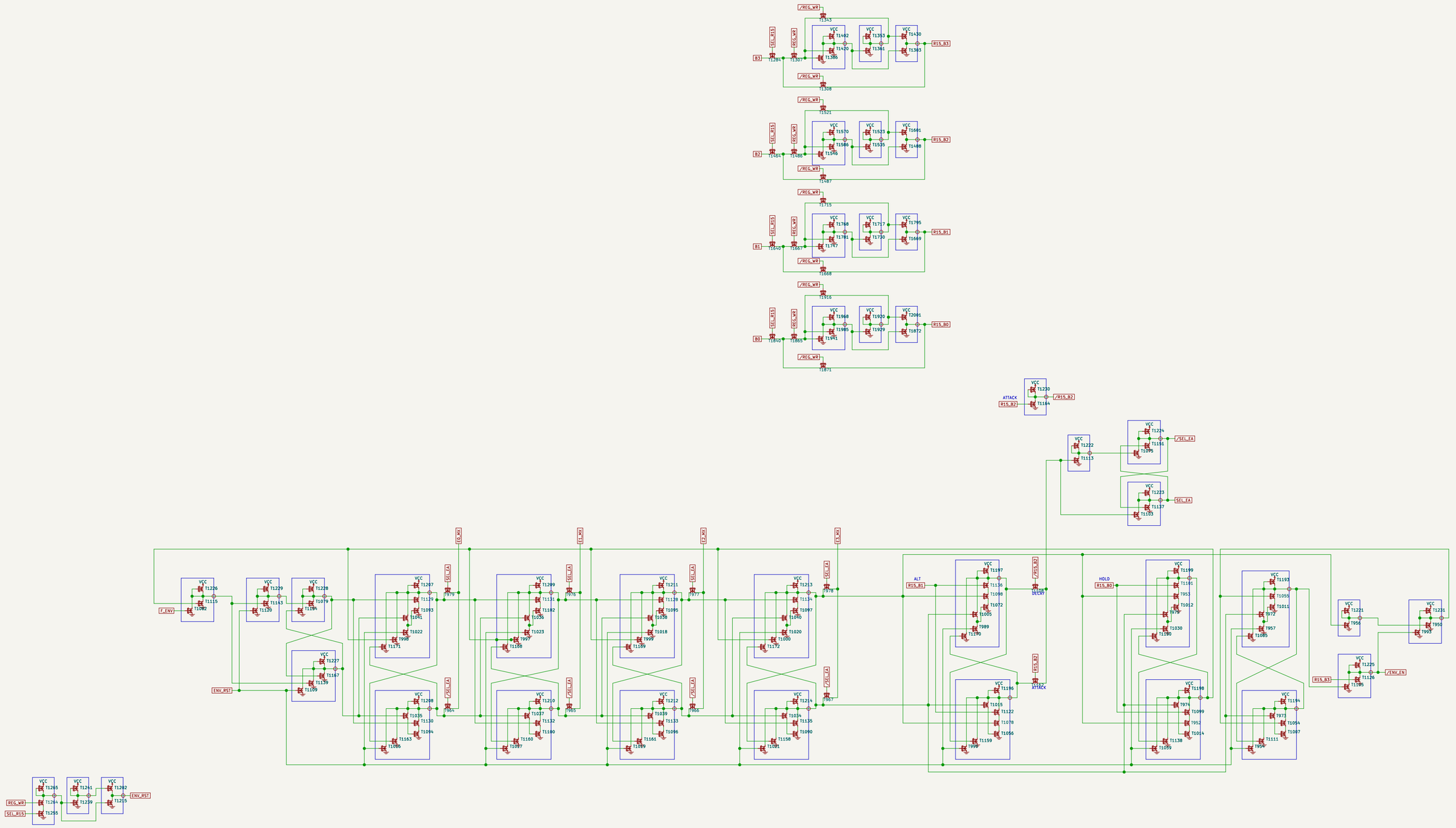


R14 (12)



Die:	General Instrument AY-3-8910 (1979)
Sheet:	3A / 16
File:	envelope_period.kicad_sch
Title:	***Envelope Period — R13 (11), R14 (12)***
Size:	A2
Date:	2026-10-09
Rev:	**1.0**
KiCad	9.0.4, 9.0.1
Id:	3/16

R15 (13) ENVELOPE MODE



Die: General Instrument AY-3-8910 (1979)		
Sheet: 15 / 16		
File: ay8910_env_s15.kicad_sch		
Title: Envelope Shape/Cycle — R15 (13)		
Size: A2	Date: 2026-10-09	**Rev: 1.0**
KiCad E.D.A. 9.0.1		Id: 1/1

B->IOA A

IOA->B L

B->IOB L

IOB->B L

PIN 14
PIN 15
PIN 16
PIN 17
PIN 18
PIN 19
PIN 20
PIN 21
PIN 6
PIN 7
PIN 8
PIN 9
PIN 10
PIN 11
PIN 12
PIN 13

Die:	General Instrument AY-3-8910 (1979)
Sheet:	16 / 16
File:	ay8910_io_s16.kicad_sch
Title:	**I/O Ports A & B - R16 (14), R17 (15)**
Size: A2	Date: 2026-10-09
KiCad E.D.A. 9.0.1	**Rev: 1.0**
	Id: 1/1